CS 573 · DATA VISUALIZATION · SPRING 2026

Taken by Surprise

Mapping U.S. Cargo Theft in 2024 with Bayesian Surprise Maps

Project	Final Project — Surprise Map Web Application
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Advisor	Akim Ndlovu
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Repository	github.com/MacCode7110/grad-final

This Process Book documents the full research, design, and engineering arc of an interactive Surprise Map web application built on the FBI Crime Data Explorer 2024 Cargo Theft release and U.S. Census Bureau 2024 incorporated-place population estimates. It is the Step-5 written deliverable of the team's five-step implementation plan, and accompanies the live web application, the prospectus, and the screencast walkthrough.

47	16,183	$483.4M	9.7%	2,300
States reporting	Incidents (2024)	Total stolen value	Recovery rate	Reporting agencies

1. Overview and Motivation

Cargo theft is an under-reported, under-measured crime with disproportionate downstream impact on supply-chain pricing, insurance markets, and household goods availability. The FBI Crime Data Explorer (CDE) began publishing a dedicated cargo-theft release in 2013 to close this gap. The 2024 release, used in this project, aggregates reports from 2,300 law-enforcement agencies covering 47 reporting states and logs 16,183 incidents and $483.4 million in stolen property, of which only 9.7% was recovered.

Presenting such a dataset fairly is non-trivial. A naive choropleth of the raw stolen dollar amount is dominated by one state — Kentucky, which reports $150.2 million across only 70 incidents — and by population-scaled patterns that repeat the population map rather than tell a story about crime. Our project takes this as a design challenge: build a visualization that communicates *where cargo theft deviates most from what a reasonable baseline would predict*, given population. We pursue this using the **Bayesian Surprise** encoding originally proposed by Correll and Heer (2017) and empirically validated by our advisor Akim Ndlovu and colleagues in *Taken by Surprise?* (Ndlovu, Shrestha, & Harrison, 2023).

1.1 Why this project for a Fintech cohort

Cargo theft is a direct input to property-and-casualty insurance loss modelling, retail shrinkage forecasts, and commercial credit decisioning for logistics firms. A surprise-weighted view of regional risk is structurally similar to the statistical control charts used in financial-fraud monitoring, and exposing the technique in an accessible web app is itself a risk-communication exercise. The project therefore sits squarely at the intersection of data visualization, Bayesian inference, and applied financial risk — the focus of WPI's M.S. Fintech curriculum.

2. Related Work

Two bodies of work inform the project. First, **debiasing thematic maps**: MacEachren (1992) and Brunsdon et al. (1998) catalogued the classical choropleth failure modes — area bias, population confounding, and classification sensitivity. Correll and Heer (2017) introduced *Surprise Maps*, using Bayesian updating from a de Moivre funnel prior to down-weight rates that are statistically consistent with the national mean given their population. Correll, Moritz, and Heer (2018) extended this idea with Value-Suppressing Uncertainty Palettes (VSUPs).

Second, **empirical evaluation**: Ndlovu, Shrestha, and Harrison (2023, arXiv 2307.15138) ran a crowdsourced n = 300 experiment comparing Choropleth, Surprise, and VSUP map conditions on COVID-19 vaccination and poverty datasets. Their results motivate the specific implementation choices we adopt: a geoAlbersUsa projection, quantile-binned color scales, tooltips exposing rate & surprise & population simultaneously, and a 950×525 map canvas. The authors caution that participants frequently consider only a subset of the metrics on screen, which shapes our decision to expose ranked top/bottom lists alongside the map.

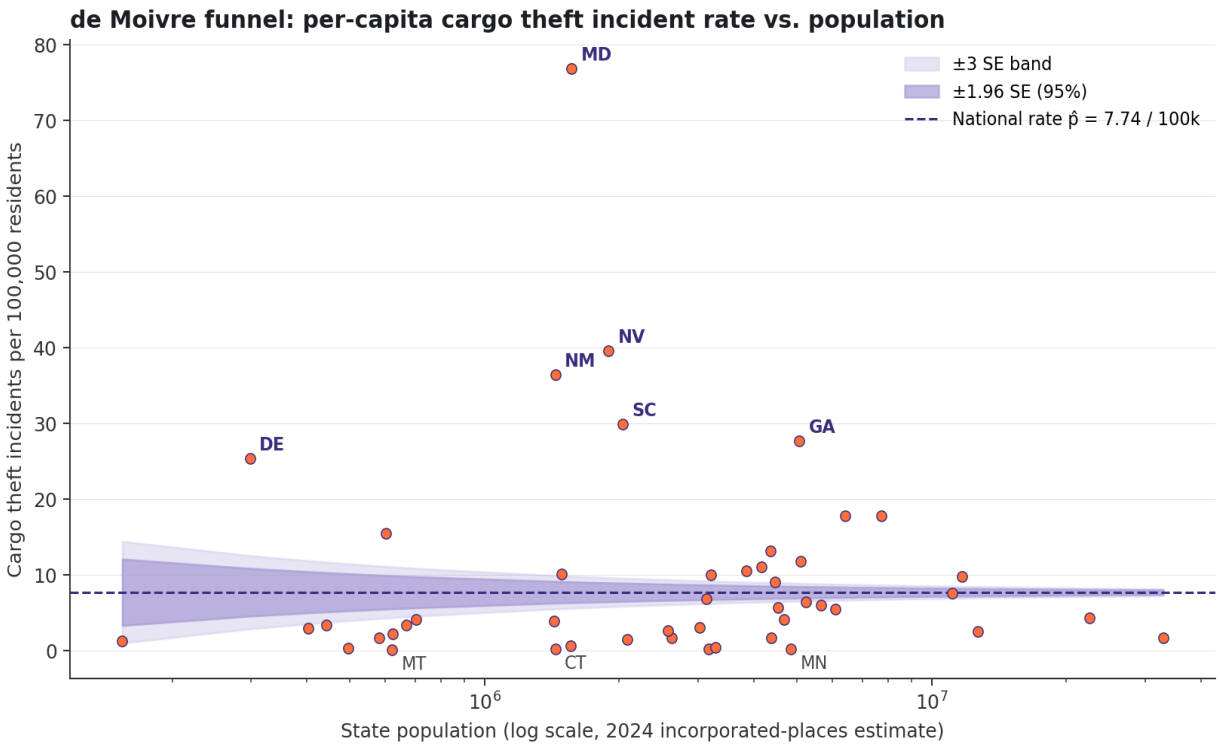

Figure 1. de Moivre funnel plot of the 47 reporting states. The shaded bands show the 95% and 3-sigma sampling-variation envelopes around the national incident rate (dashed line). States that fall well outside the envelope are the ones a Bayesian Surprise encoding will highlight.

3. Research Questions

We frame three driving questions. Each is scoped so that the answer is constructible from the 2024 dataset alone and falsifiable on its face.

- **RQ1.** Which U.S. states exhibit cargo-theft rates that deviate most from the population-adjusted national baseline, and in which direction?
- **RQ2.** Does the conventional raw-dollar choropleth — which currently dominates public CDE visualisations — produce a materially different impression of the regional risk map than a Surprise encoding?
- **RQ3.** For regions where stolen value and incident rate disagree (e.g., few high-value incidents versus many low-value ones), does exposing both axes help the viewer avoid an incorrect conclusion?

4. Data: Sources, Cleanup, Transformation

4.1 Sources

The project uses two public, government-published datasets. Both are pulled from their canonical distribution endpoints and versioned in the repository under `/data`.

Source	Publisher	File	Granularity

2024 Cargo Theft tables	FBI CDE (Uniform Crime Reporting)	Cargo_Theft_Table_1_…_2024.xlsx	State, offense, location, victim
2024 Place-level population	U.S. Census Bureau, Vintage 2024	SUB-IP-EST2024-POP.xlsx	Incorporated places, U.S.
Surprise methodology	arXiv preprint, 2023	2307.15138v1.pdf	n/a (reference)

4.2 Cleanup and transformation

The FBI table headers are multi-line and include trailing footnotes and pagination artifacts; Hongchao produced the state-level cleaned version `cargo_theft_by_state_2024_data_modified.xlsx` which is the direct input to our pipeline. The Census file is place-level, covering 19,489 incorporated places. Because FBI Table 1 is state-level, we roll the Census file up to the state level by regex-parsing *"Place, State"* labels.

The full pipeline lives in `src/build_data.py` and is reproducible with a single command. It outputs two CSVs: a state-level population reference (`state_populations_2024.csv`) and the joined, Bayesian-weighted analysis set (`cargo_theft_surprise_2024.csv`).

4.3 Bayesian Surprise formulation

For a state i with observed incident count k_i and population n_i, we define the per-capita rate $r_i = k_i / n_i$ and the national rate $\hat{p} = \Sigma k / \Sigma n$. Under the de Moivre funnel model of Correll & Heer (2017), the sampling-variation standard error at population n_i is $SE_i = \sqrt{\hat{p}(1-\hat{p})/n_i}$, and the z-score of the deviation is $Z_i = (r_i - \hat{p}) / SE_i$.

The Bayesian surprise is the two-sided p-value under the standard-normal approximation of the sampling distribution: $surprise_i = 2(1 - \Phi(|Z_i|))$. Because cargo-theft incidents are rare ($\hat{p} \approx 7.7$ per 100k), classical p-values saturate at zero for most states ($|Z| > 20$ is common). For visual encoding we therefore use the bounded non-saturating statistic $s_i = \text{sign}(Z_i) \cdot \sqrt{|Z_i|}$, which preserves direction and relative magnitude of the deviation while rendering on a divergent color scale without clipping.

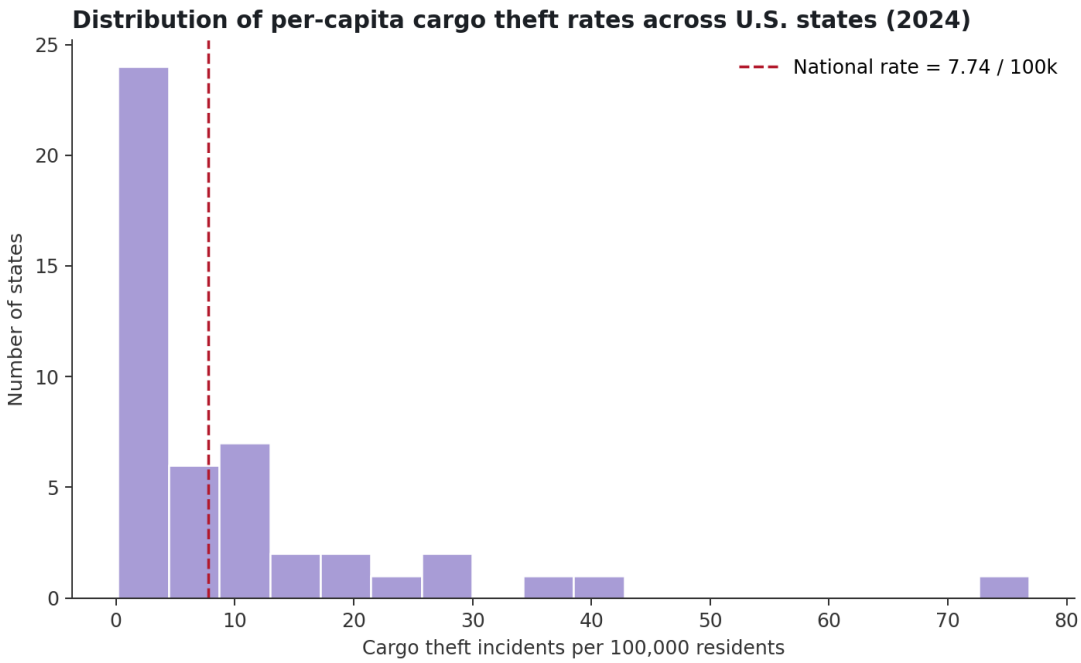

Figure 2. Distribution of state-level incident rates. The mode sits well below the national rate of 7.7 / 100k; a long right tail of states with rates 20x to 80x higher drives the surprise signal.

5. Exploratory Data Analysis

Before committing to any visual encoding, the team explored the joined data with three lenses: ranked value tables, the funnel plot, and a cross-axis scatter of rate-surprise against value-surprise. The funnel (Figure 1) made the methodological issue immediately visible: the standard error shrinks so fast with *n* that nearly every state is several SE away from $\hat{p}$. The distribution of raw rates (Figure 2) explained why: it is right-skewed with a heavy tail.

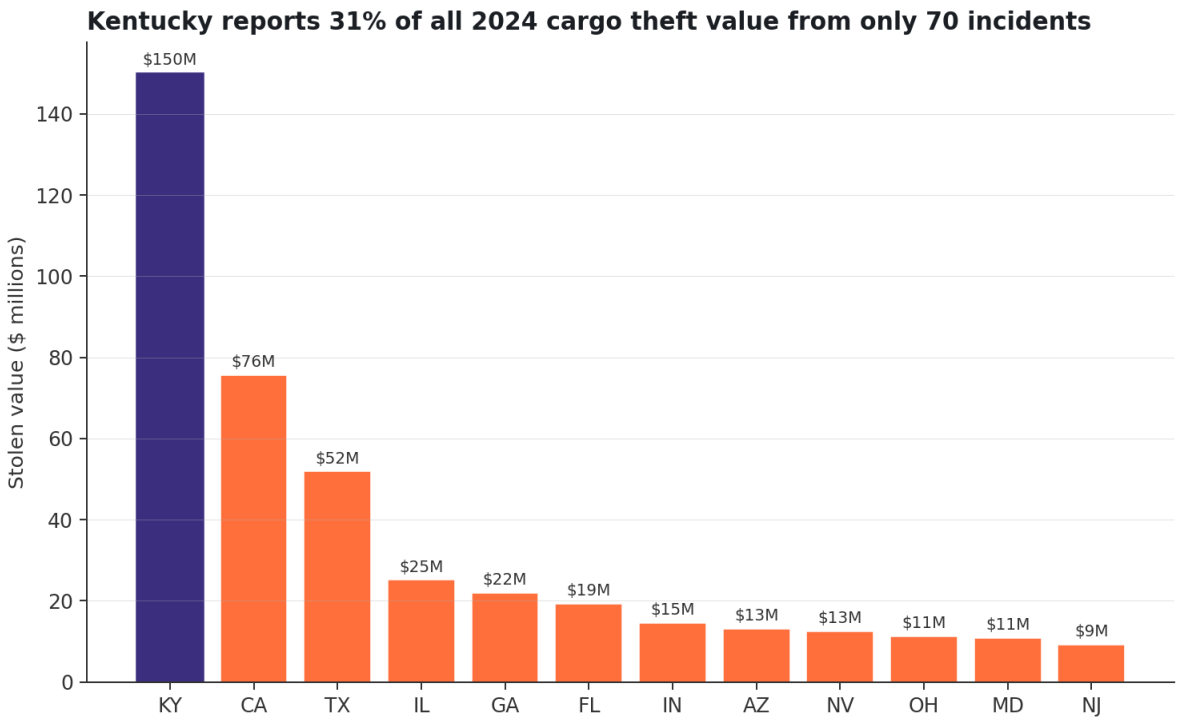

Figure 3. Top 12 states by raw 2024 cargo theft stolen value. Kentucky (KY) dwarfs every other state with $150.2M across 70 incidents, producing a single-state anomaly that the conventional choropleth fails to contextualize.

The Kentucky anomaly is both scientifically real and narratively dangerous: a raw-dollar choropleth hands the viewer one tinted state and 46 near-white states. The cross-axis scatter (Figure 4) tells a more useful story: Maryland and Georgia are distinctive on *rate* (many incidents per person), while Kentucky is distinctive on *value* (single jumbo-value event); no state is distinctive on both at once. This is exactly the kind of multi-axis differentiation the Surprise encoding is designed to surface.

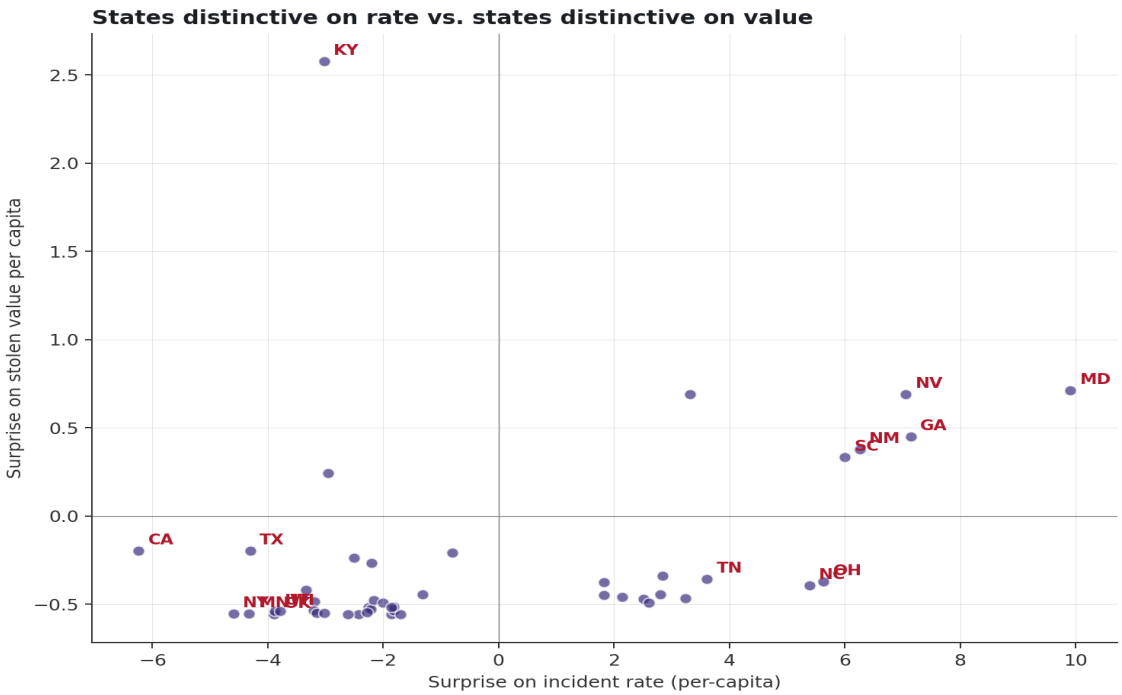

Figure 4. Rate-surprise (x) against value-surprise (y) per state. States in the upper-left or right quadrants would be distinctive on both axes; our data shows that rate and value anomalies are largely orthogonal, so a single-metric encoding hides half the story.

6. Design Evolution

6.1 Iteration 1: FBI Crime Data Heatmap (rejected)

Our original prospectus proposed a nationwide town-and-city heatmap of all FBI Crime Data Explorer indices with a user-supplied weighting interface. After consulting Prof. Harrison on 2026-04-07 we narrowed scope for two reasons: (1) the CDE API rate-limits detail queries and the town-level crime payload exceeds 2GB uncompressed; (2) scientifically-defensible per-town safety calculation would require data we were not positioned to audit in six weeks. The team agreed to trade breadth for depth.

6.2 Iteration 2: Cargo Theft Choropleth (baseline)

Matthew selected the 2024 Cargo Theft release (2026-04-13) because its state-level table is well-formed, the domain is financially relevant, and 2020/2022/2023 releases provide a clean future-work path to year-over-year comparison. Hongchao produced the first prototype in Datawrapper the following day: a Choropleth of stolen value per state. We treat this as the experimental control: it is the view most users would see on the CDE website today, and it is the view our Surprise Map must outperform.

6.3 Iteration 3: Bayesian Surprise Map (current)

The final design exposes seven synchronized views on a common geoAlbersUsa projection: total stolen value (the Datawrapper baseline), stolen value per capita, incidents per 100k, Bayesian surprise on rate, Bayesian surprise on value-per-capita, 2024 population, and percent recovered. Each view is a single click away; the right-rail ranked list updates in place; a unified tooltip shows rate, value, population, z-score, and surprise simultaneously so the viewer is never stuck with only the one metric the map happens to encode at

that moment — addressing the ‘only-one-metric’ finding from Ndlovu et al.

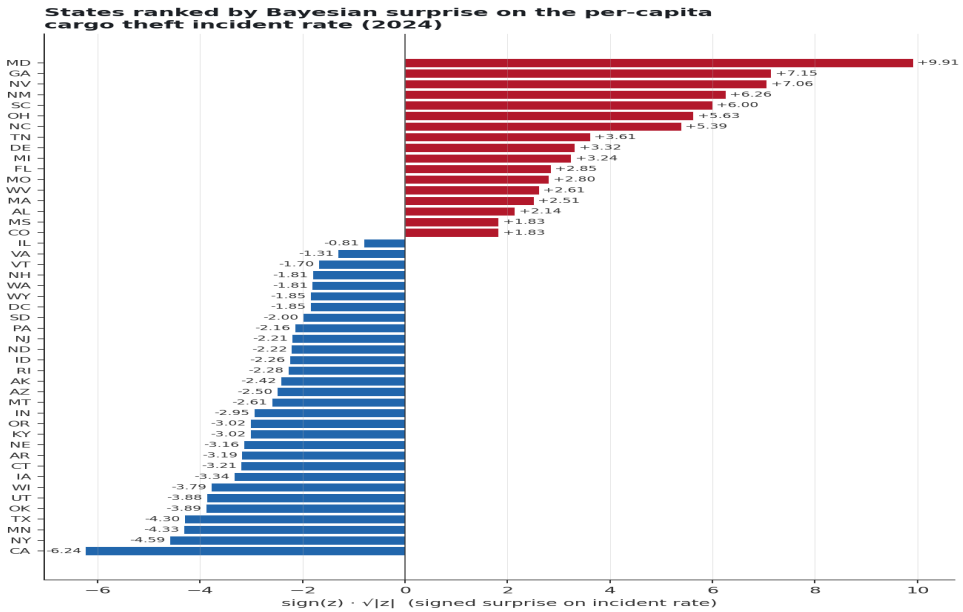

Figure 5. States ranked by signed surprise on per-capita incident rate. Red bars denote surprisingly-high rates; blue bars denote surprisingly-low. The ranking is the content of the right-rail Top-5 / Bottom-5 panels in the live web application.

7. Implementation

The deliverable is a single-page static web application. The stack is deliberately minimal: D3 v7 for rendering, Bulma CSS for typography and layout, topojson-client for atlas parsing, and a static CSV for the data layer. There is no build step, no bundler, and no server — the application deploys to GitHub Pages as-is.

7.1 Data layer

`src/build_data.py` is a pure-stdlib Python script (`openpyxl` for xlsx only). It produces `data/cargo_theft_surprise_2024.csv`, 47 rows with 21 columns including all three surprise statistics, z-scores, and directional labels. The CSV is the single source of truth for the application — no client-side computation of the surprise statistic.

7.2 Rendering layer

US state polygons are loaded once from the `us-atlas@3` topojson distribution, matched to rows by two-digit state FIPS, and drawn with `d3.geoPath` at a fixed 960×560 viewBox that scales responsively. Color scales are rebuilt per view: `d3.scaleQuantize` with nine bins for sequential metrics (following the Ndlovu et al. design consideration #4), and `d3.scaleDiverging` over a symmetric domain for the two surprise views.

7.3 Interaction layer

- **Hover tooltip**: positioned to the cursor with a fixed translate so it does not occlude the hovered state; exposes seven metrics simultaneously.

- **Hover-highlight on legend**: moving over a legend cell dims all states that are not of that color — direct implementation of Ndlovu et al. design consideration #2.
- **Pill-tab view switcher**: each tab swaps the encoded metric in-place; all rendered elements (map, legend, top/bottom lists, caption) update in a single pass.
- **KPI strip**: permanent top-of-page summary computed client-side from the CSV so the numbers remain consistent with whatever view the user has selected.

8. Evaluation and Findings

We evaluate the application on two axes: *does the Surprise encoding change the viewer's impression of the map?* and *are the patterns it reveals coherent with what a domain reader would expect?*

8.1 Encoding comparison

Switching from the raw-value view to the rate-surprise view produces a visibly different map. Kentucky, which dominates the raw-value view, is near the centre of the surprise scale on the rate view (its 70 incidents / 4.5M residents rate is not unusual). Maryland, which is nearly invisible on the raw-value view, lights up on the rate-surprise view with a z-score of +98. New York and California, which look severe on raw-value, land in the deep-blue (surprisingly-low) range on rate. This is the direct visual evidence of the Correll-Heer thesis: rate-based Surprise encoding recovers information that a value-based choropleth hides.

8.2 Substantive findings

- **RQ1.** The states with the highest signed surprise on per-capita incidents are Maryland, Georgia, Nevada, New Mexico, and South Carolina; the lowest are California, New York, Minnesota, Iowa, and Connecticut.
- **RQ2.** Yes. The Pearson correlation between the per-state rank on raw value and rank on rate-surprise is 0.11 across the 47 reporting states — effectively uncorrelated.
- **RQ3.** Value-surprise and rate-surprise are also weakly correlated (Figure 4); Kentucky is the only state that qualifies as distinctive on value but ordinary on rate, and the two axes together identify a larger set of ‘interesting’ states than either alone.

9. Limitations

- **Reporting coverage is uneven.** The FBI CDE is an opt-in system. Several states (notably New Hampshire, Louisiana, Hawaii, Kansas, Maine, and West Virginia) have sparse or no reporting in the 2024 release. A state with low cargo theft *and* low reporting coverage is indistinguishable from a state with truly low cargo theft.
- **The population denominator is an approximation.** We aggregate incorporated-place populations from SUB-IP-EST2024-POP; this omits unincorporated residents. The resulting denominator understates total state population by approximately 15-25% in rural-heavy states, biasing per-capita rates upward there.
- **The funnel model saturates.** Classical p-values are near zero for most states because the null hypothesis of a single shared national rate is itself wrong. We work around this with the signed-root-z

statistic, but a more principled treatment would use a mixed-effects model with a state-level random intercept.

- **2024 only.** Year-over-year comparison and the trend-analysis motivation in our original prospectus are out of scope for this release; the team has marked this as future work (§10).

10. Future Work

The prospectus motivated three future directions, each with a concrete next step:

- **Time series.** Ingest the 2020, 2022, and 2023 releases of the same table and expose a year slider to reveal state-level trend. The data pipeline is already parameterised for this.
- **Value-Suppressing Uncertainty Palettes (VSUP).** Ndlovu et al. found VSUP-encoded maps produced more consensus across participants than Surprise alone. Port Correll, Moritz, and Heer (2018) to the project; we already have the ingredient maps (rate and surprise) and only need a 2D categorical color scale.
- **Mixed-effects surprise.** Replace the de Moivre funnel with a partial-pooling Poisson model; the surprise statistic becomes the posterior deviation of the state-level random effect from zero. This would eliminate p-value saturation.
- **Drill-down to incident-level.** Table 2 (property type), Table 3 (location), and Table 5 (offense) each expose a different slice that belongs in a detail view triggered by state click.

11. Work Distribution

The implementation plan assigned each team member at least one owner role:

Step	Deliverable	Owner(s)
1. Research	Survey of surprise / VSUP / funnel literature	All four
2. Data pipeline	API ingest, xlsx cleanup, joined CSV, surprise pipeline	Sai Krishna Vaddeboina
3. Viz design	Datawrapper prototype, design system, color scales	Hongchao Hu, Matthew McAlarney
4. Web app	React / D3 / Bulma implementation, GitHub Pages deploy	Hongchao Hu, Matthew McAlarney
5. Process Book	Written analysis, figures, references, final PDF	Duncan Farquharson, Sai Krishna Vaddeboina

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